

Chapter 4

Two Dimensional Elastostatics

4.1 Introduction

This chapter deals with the applications of boundary elements to solve two dimensional elastostatics problems. The basic equations of elasticity are reviewed, first pointing out that boundary elements for these cases are based on the plane strain approach but can be extended to plane stress if the elastic coefficients are replaced by the corresponding equivalent values, as will be seen below.

The boundary element formulation of elastostatics is substantially different from the one for potential problems as its unknowns are in vector rather than scalar form. This means that at a particular point there are two (or three in three dimensions) components of displacements or tractions, rather than one of each as discussed in Chapter 3. The associated fundamental solution also represents a point load acting in a given direction. All this implies that the boundary element formulation for elastostatics is more complex than the one described in Chapter 2 for potential problems although it involves the same basic steps.

It seems appropriate to start by developing the necessary theory and then the computer code corresponding to the simplest case, i.e. the one with constant elements. After this the development of a quadrilateral element and associated code is described, without presenting the linear case in detail. This approach has been preferred as the linear case has already been discussed in Chapter 2 for potential problems and is comparatively simple to extend the formulation to elastostatics. The quadratic elasticity code is also more interesting from the point of view of applications, as many engineering problems are difficult to solve accurately using constant elements and the linear codes do not converge rapidly for cases involving bending, for instance. Hence while constant or linear elements can be satisfactorily applied for many potential problems they – particularly the constant ones – are generally not sufficiently accurate for stress analysis applications.

The chapter presents a series of examples describing the excellent results which can be obtained by using boundary elements in elastostatics. This chapter in conjunction with the previous one attempts to be an introduction to the use of boundary elements for stress analysis. For further developments and more complex applications the reader is advised to consult the relevant publications listed in Appendix C.

4.2 Plate Stretching – Plane Strain Problems

Two dimensional problems are divided into two types, plate stretching (sometimes

also called plane stress) and plane strain problems, depending on how the solid is restrained in the direction perpendicular to the plane under study.

To understand the difference between these two states consider the prismatic homogeneous solid shown in figure 4.1. The end surfaces are defined by the planes at $x_3 = \pm h/2$, and the cylindrical surface by $x_1 = x_1(\Gamma)$, $x_2 = x_2(\Gamma)$, where Γ is the arc length along the boundary curve.

Plate stretching The basic assumptions for plate stretching are

- (i) that the body is thin, i.e. h is small by comparison with the representative dimensions along x_1 or x_2 ,
- (ii) there are no tractions acting at the end surfaces, i.e. at $x_3 = \pm h/2$, $p_j = 0$,
- (iii) the body forces are acting on $x_1 - x_2$ planes and independent of x_3 , i.e. $b_3 = 0$ and b_1, b_2 are functions of x_1, x_2 only,
- (iv) the forces acting on the cylindrical body are planar and independent of x_3 , i.e. $p_3 = 0$ and p_1, p_2 are functions of x_1 and x_2 .

Under these assumptions it is assumed that σ_{33} , σ_{31} and σ_{32} are all small in comparison with $\sigma_{11}, \sigma_{22}, \sigma_{12}$ and that the variation of the latter with respect to x_3 is negligible. Hence one assumes:

$$\sigma_{33} = \sigma_{31} = \sigma_{32} = 0 \quad (4.1)$$

and σ_{11} , σ_{22} and σ_{12} are functions of x_1, x_2 only.

$$\sigma_{11}(x_1, x_2); \quad \sigma_{22}(x_1, x_2); \quad \sigma_{12}(x_1, x_2)$$

It should be noticed however that although these assumptions are reasonable in engineering practice they are only approximate as they violate the compatibility equations.

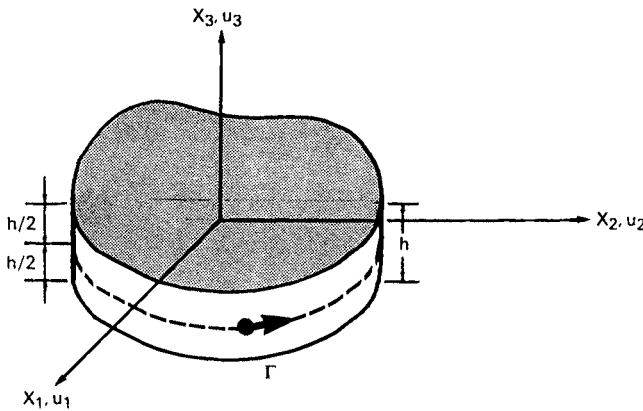


Figure 4.1 Prismatic solid

Equation (4.1) is written in function of displacements and one can have an alternative statement, i.e. the displacement components are functions of x_1, x_2 only, i.e.

$$u_1 = u_1(x_1, x_2); \quad u_2 = u_2(x_1, x_2) \quad (4.2)$$

Notice that $u_3 \neq 0$ and the ϵ_{33} strains can be determined in function of the σ_{11} , σ_{22} , σ_{12} stresses.

Plane Strain This case usually represents the behaviour of long structures such as tunnels and for this the displacements in the normal direction can be assumed to be zero (i.e. on the end faces).

The the plane strain assumptions are

- (ii) The end faces displacements u_3 are zero as they are restrained to move normally because the thickness is large in comparison with the representative dimensions in x_1, x_2 directions.
- (ii) The body and surface forces acting on the cylindrical surface have no x_3 component and are independent of x_3 .

In this case in addition to $u_3 = 0$, one can assume that the in-plane displacements u_1 and u_2 are independent of x_3 . Hence

$$u_1 = u_1(x_1, x_2); \quad u_2 = u_2(x_1, x_2); \quad u_3 = 0 \quad (4.3)$$

This means that some of the resulting strains will also be zero, i.e.

$$\epsilon_{33} = \epsilon_{31} = \epsilon_{32} = 0 \quad (4.4)$$

and the others independent of x_3 , i.e. $\epsilon_{11}(x_1, x_2)$, $\epsilon_{22}(x_1, x_2)$, $\epsilon_{12}(x_1, x_2)$.

For this case $\sigma_{33} \neq 0$ and can be determined from the value of the other components.

Constitutive Relations

One can now expand the three dimensional stress-strain relationships for an isotropic body in terms of the non-zero components for plane strain or plane stress, i.e.

$$\begin{aligned} \epsilon_{11} &= \frac{1}{E} (\sigma_{11} - \nu \sigma_{22} - \nu \sigma_{33}) \\ \epsilon_{22} &= \frac{1}{E} (-\nu \sigma_{11} + \sigma_{22} - \nu \sigma_{33}) \\ \epsilon_{33} &= \frac{1}{E} (-\nu \sigma_{11} - \nu \sigma_{22} + \sigma_{33}) \\ \epsilon_{12} &= \frac{1}{2G} \sigma_{12} \end{aligned} \quad (4.5)$$

The relationships for plate stretching can be obtained by setting $\sigma_{33} \equiv 0$ in (4.5) and considering only the planar components ε_{11} , ε_{22} and ε_{12} , i.e.

$$\begin{aligned}\varepsilon_{11} &= \frac{1}{E} (\sigma_{11} - \nu \sigma_{22}) \\ \varepsilon_{22} &= \frac{1}{E} (-\nu \sigma_{11} + \sigma_{22}) \\ \varepsilon_{12} &= \frac{1}{2G} \sigma_{12}\end{aligned}\tag{4.6}$$

The value of ε_{33} can be obtained a posteriori from the third relationship in (4.5), i.e.

$$\varepsilon_{33} = \frac{1}{E} (-\nu \sigma_{11} - \nu \sigma_{22})\tag{4.7}$$

Equations (4.6) can be inverted to produce,

$$\begin{aligned}\sigma_{11} &= \frac{E}{1 - \nu^2} (\varepsilon_{11} + \nu \varepsilon_{22}) \\ \sigma_{22} &= \frac{E}{1 - \nu^2} (\nu \varepsilon_{11} + \varepsilon_{22}) \\ \sigma_{12} &= 2G \varepsilon_{12}\end{aligned}\tag{4.8}$$

The plane strain equations can be found by first eliminating ε_{33} from (4.5), i.e.

$$\varepsilon_{33} = \frac{1}{E} (-\nu \sigma_{11} - \nu \sigma_{22} + \sigma_{33}) \equiv 0\tag{4.9}$$

which gives

$$\sigma_{33} = \nu (\sigma_{11} + \sigma_{22})\tag{4.10}$$

Substituting for σ_{33} in equation (4.5) one obtains the following expressions,

$$\begin{aligned}\varepsilon_{11} &= \frac{1}{E} [(1 - \nu^2) \sigma_{11} - \nu(1 + \nu) \sigma_{22}] \\ \varepsilon_{22} &= \frac{1}{E} [-\nu(1 + \nu) \sigma_{11} + (1 - \nu^2) \sigma_{22}] \\ \varepsilon_{12} &= \frac{1}{2G} \sigma_{12}\end{aligned}\tag{4.11}$$

The inverse of these expressions is

$$\begin{aligned}\sigma_{11} &= \frac{E}{(1+\nu)(1-2\nu)} [(1-\nu)\varepsilon_{11} + \nu\varepsilon_{22}] \\ \sigma_{22} &= \frac{E}{(1+\nu)(1-2\nu)} [\nu\varepsilon_{11} + (1-\nu)\varepsilon_{22}] \\ \sigma_{12} &= 2G\varepsilon_{12}\end{aligned}\tag{4.12}$$

The value of ε_{33} if required can be obtained from (4.10).

Notice that it is sometimes more convenient to express the first two relationships above using Lamé's constant $\lambda = \nu E / [(1+\nu)(1-2\nu)]$, i.e.

$$\begin{aligned}\sigma_{11} &= \frac{\lambda}{\nu} [(1-\nu)\varepsilon_{11} + \nu\varepsilon_{22}] \\ \sigma_{22} &= \frac{\lambda}{\nu} [(1-\nu)\varepsilon_{22} + \nu\varepsilon_{11}]\end{aligned}\tag{4.13}$$

One can pass from equations (4.8) for plate stretching to equations (4.12) for plane strain simply by replacing E and ν in the first equation by two equivalent values, E' and ν' , such that

$$E' = \frac{E}{1-\nu^2}; \quad \nu' = \frac{\nu}{1-\nu}\tag{4.14}$$

where the value of G remains the same.

This interesting relationship means that one can implement a plate stretching program and by transforming the elastic constant data in accordance to (4.14) solve also a plane strain problem. This is done in finite element analysis. Conversely in boundary elements one works with plane strain problems as the fundamental solution is known for this type of problem and then plate stretching problems can be solved using the inverse relationship to (4.14), i.e.

$$E' = (1-\nu^2)E; \quad \nu' = \frac{\nu}{1+\nu}\tag{4.15}$$

4.3 Boundary Element Formulation

The basic relationships for boundary elements in elastostatics have been developed in Chapter 3. In what follows some of the corresponding equations will be reviewed and specialized for two dimensional cases with a view to applying them in two computer codes, one using constant element and the other quadratic elements, the latter more appropriate for elasticity problems. The development of an

elastostatics linear code if required, is left to the reader as it follows similar lines as the linear code for potential problems.

One can start with the integral expression (3.79), i.e.

$$c_{ik}^i u_k^i + \int_{\Gamma} u_k p_{ik}^* d\Gamma = \int_{\Gamma} p_k u_{ik}^* d\Gamma + \int_{\Omega} b_k u_{ik}^* d\Omega \quad (4.16)$$

where the fundamental solution has been assumed to satisfy the following equation,

$$\sigma_{jk,j}^* + \Delta^i = 0 \quad (4.17)$$

Equation (4.16) applies for points on the boundary or internal points (with $c_{ik}^i = \delta_{ik}$). Smooth surfaces give $c_{ik}^i = \frac{1}{2}\delta_{ik}$ and corners produce a different type of c_{ik}^i array as will be discussed shortly.

The fundamental solution for an isotropic material in plane strain has been given in Chapter 3 but will be repeated here for completeness.

$$\begin{aligned} u_{ik}^* &= \frac{1}{8\pi G(1-\nu)} \left[(3-4\nu) \ln\left(\frac{1}{r}\right) \delta_{ik} + \frac{\partial r}{\partial x_i} \frac{\partial r}{\partial x_k} \right] \\ p_{ik}^* &= -\frac{1}{4\pi(1-\nu)r} \left[\frac{\partial r}{\partial n} \left\{ (1-2\nu)\delta_{kl} + 2 \frac{\partial r}{\partial x_k} \frac{\partial r}{\partial x_l} \right\} \right. \\ &\quad \left. - (1-2\nu) \left(\frac{\partial r}{\partial x_l} n_k - \frac{\partial r}{\partial x_k} n_l \right) \right] \end{aligned} \quad (4.18)$$

p_{ik}^* and u_{ik}^* represent the tractions and displacements in the k direction due to a unit load in the l direction acting at 'i'.

One can now write equation (4.16) in matrix form by defining the following arrays. The fundamental solution components can be written as two 2×2 matrices with elements u_{ik}^* and p_{ik}^* , i.e.

$$\mathbf{u}^* = \begin{bmatrix} u_{11}^* & u_{12}^* \\ u_{21}^* & u_{22}^* \end{bmatrix}; \quad \mathbf{p}^* = \begin{bmatrix} p_{11}^* & p_{12}^* \\ p_{21}^* & p_{22}^* \end{bmatrix} \quad (4.19)$$

The displacement, tractions and body forces vectors are

$$\mathbf{u} = \begin{Bmatrix} u_1 \\ u_2 \end{Bmatrix}; \quad \mathbf{p} = \begin{Bmatrix} p_1 \\ p_2 \end{Bmatrix}; \quad \mathbf{b} = \begin{Bmatrix} b_1 \\ b_2 \end{Bmatrix} \quad (4.20)$$

Hence the basic equation becomes,

$$\mathbf{c}^i \mathbf{u}^i + \int_{\Gamma} \mathbf{p}^* \mathbf{u} d\Gamma = \int_{\Gamma} \mathbf{u}^* \mathbf{p} d\Gamma + \int_{\Omega} \mathbf{u}^* \mathbf{b} d\Omega \quad (4.21)$$

where the $\mathbf{u}^i$ defines the displacements at the - internal or boundary - point i

where the load is applied. $\mathbf{c}^i$ is a 2×2 array of constant which values depend on the type of point under consideration. If 'i' is an internal point

$$\mathbf{c}^i = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \quad (4.22)$$

If 'i' is a boundary point on a smooth surface then,

$$\mathbf{c}^i = \begin{bmatrix} \frac{1}{2} & 0 \\ 0 & \frac{1}{2} \end{bmatrix} \quad (4.23)$$

If 'i' is a corner, we will have

$$\mathbf{c}^i = \begin{bmatrix} c_{11} & c_{12} \\ c_{21} & c_{22} \end{bmatrix} \quad (4.24)$$

where the c_{ik} values will depend on the type of corner under consideration as we will see shortly.

4.4 Constant Element Formulation

Consider now that the surface of the boundary under study is discretized using constant elements (figure 4.2). This implies that the values of $\mathbf{u}$ and $\mathbf{p}$ are assumed to be constant on each element and equal to the value at the mid-node of the element. One can also discretize the interiors of the domain in a number of cells which are required for the integration of the body force term in (4.21). These cells are used only for numerical integration of the body force terms and in certain cases – as seen in Chapter 3 – they can be taken to the boundary. Consider here that the body Ω is discretized into N boundary elements and M internal cells, hence formula (4.21) can be written as,

$$\mathbf{c}^i \mathbf{u}^i + \sum_{j=1}^N \left\{ \int_{\Gamma_j} \mathbf{p}^* d\Gamma \right\} \mathbf{u}^j = \sum_{j=1}^N \left\{ \int_{\Gamma_j} \mathbf{u}^* d\Gamma \right\} \mathbf{p}^j + \sum_{s=1}^M \left\{ \int_{\Omega_s} \mathbf{u}^* \mathbf{b} d\Omega \right\} \quad (4.25)$$

The above equation corresponds to the particular node 'i' where the unit forces are assumed to be acting.

Notice that terms such as $\int_{\Gamma_j} \mathbf{u} d\Gamma$ and $\int_{\Gamma_j} \mathbf{p} d\Gamma$ relate node 'i' to the element or node 'j'. They produce a type of influence coefficient. After integration the integrals produce two 2×2 submatrices called $\hat{\mathbf{H}}^{ij}$ and $\mathbf{G}^{ij}$. Numerical integration of the body forces term can be carried out as follows,

$$\sum_{s=1}^M \left\{ \int_{\Omega_s} \mathbf{u}^* \mathbf{b} d\Omega \right\} = \sum_{s=1}^M \left\{ \sum_{p=1}^I (\mathbf{u}^* \mathbf{b})_p w_p \right\} \Omega_s = \sum_{s=1}^M \mathbf{B}^{is} \quad (4.26)$$

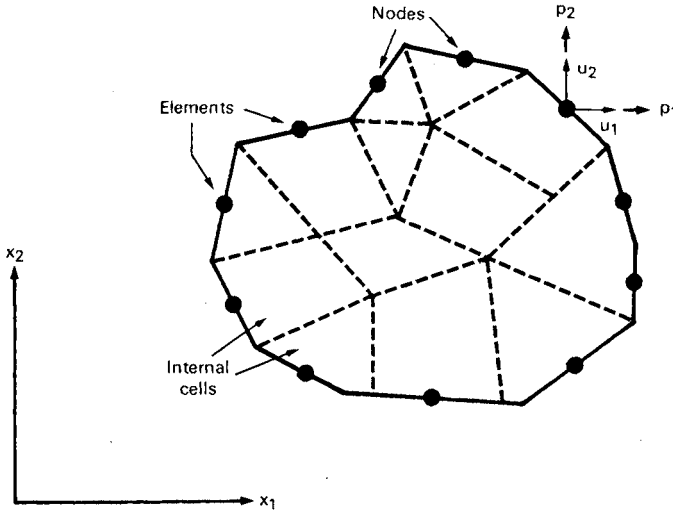


Figure 4.2 Two-dimensional body divided into boundary elements and internal cells

Notice that this produces two components of $\mathbf{B}^{is}$, i.e. B_1^{is} and B_2^{is} after the numerical integration has taken place. w_p are the weighting coefficients and Ω_s the area of the cell under consideration. The function $(\mathbf{u}^*\mathbf{b})$ has to be calculated at the p integration points, where p varies from 1 to r .

Equation (4.25) can now be written

$$\mathbf{c}^i \mathbf{u}^i + \sum_{j=1}^N \hat{\mathbf{H}}^{ij} \mathbf{u}^j = \sum_{j=1}^N \mathbf{G}^{ij} \mathbf{p}^j + \sum_{s=1}^M \mathbf{B}^{is} \quad (4.27)$$

This equation relates the values of $\mathbf{u}$ at node 'i' with the values of $\mathbf{u}$'s and $\mathbf{p}$'s at all the nodes on the boundary, including 'i'. Notice that in this case – smooth boundary – $\mathbf{c}^i$ is a 2×2 matrix with only $\frac{1}{2}$ on the diagonal.

Equation (4.27) can be written in a more compact manner if we define

$$\begin{aligned} \mathbf{H}^{ij} &= \hat{\mathbf{H}}^{ij} & \text{when } i \neq j \\ \mathbf{H}^{ij} &= \hat{\mathbf{H}}^{ij} + \mathbf{c}^i & \text{when } i = j \end{aligned} \quad (4.28)$$

Formula (4.27) has then the following form

$$\sum_{j=1}^N \mathbf{H}^{ij} \mathbf{u}^j = \sum_{j=1}^N \mathbf{G}^{ij} \mathbf{p}^j + \sum_{s=1}^M \mathbf{B}^{is} \quad (4.29)$$

If one applies (4.29) to all boundary points the result can also be written in matrix form, i.e.

$$\mathbf{H}\mathbf{U} = \mathbf{G}\mathbf{P} + \mathbf{B} \quad (4.30)$$

where $\mathbf{H}$ and $\mathbf{G}$ are $2N \times 2N$ matrices (N number of boundary nodes).

Equation (4.30) has to be rearranged when applying the boundary conditions. The process consists of moving to the left hand side all columns multiplied by an unknown and accumulating on the right hand side vector $\mathbf{F}$ all the values obtained by multiplying the known boundary conditions by the terms in the corresponding columns. This produces the following system of equations:

$$\mathbf{AX} = \mathbf{F} + \mathbf{B} \quad (4.31)$$

The vector $\mathbf{X}$ represents all unknowns – displacements or tractions – in the problem. Once (4.31) is solved all boundary values are found.

Results at Internal Points

Once the values of displacements and tractions are known on the boundary it is possible to calculate the displacements and stresses at any interior point. The displacements are given by formula (4.25) with $\mathbf{c}^i = \mathbf{I}$ ($\mathbf{I}$ is unit diagonal matrix), i.e.

$$\mathbf{u}^i = \int_{\Gamma} \mathbf{u}^* \mathbf{p} \, d\Gamma - \int_{\Gamma} \mathbf{p}^* \mathbf{u} \, d\Gamma + \int_{\Omega} \mathbf{u}^* \mathbf{b} \, d\Omega \quad (4.32)$$

This expression can be discretized

$$\mathbf{u}^i = \sum_{j=1}^N \left\{ \int_{\Gamma_j} \mathbf{u}^* \, d\Gamma \right\} \mathbf{p}^j - \sum_{j=1}^N \left\{ \int_{\Gamma_j} \mathbf{p}^* \, d\Gamma \right\} \mathbf{u}^j + \sum_{s=1}^M \left\{ \int_{\Omega_s} \mathbf{u}^* \mathbf{b} \, d\Omega \right\} \quad (4.33)$$

and then integrated numerically, analytically or by a combination of both techniques.

The internal stresses can be found using the formula (3.109), i.e.

$$\sigma_{kl} = \int_{\Gamma} \mathbf{D}_{kl} \mathbf{p} \, d\Gamma - \int_{\Gamma} \mathbf{S}_{kl} \mathbf{u} \, d\Gamma + \int_{\Omega} \mathbf{D}_{kl} \mathbf{b} \, d\Omega \quad (4.34)$$

where

$$\mathbf{D}_{kl} = [D_{1kl}, D_{2kl}]$$

$$\mathbf{S}_{kl} = [S_{1kl}, S_{2kl}]$$

Formula (4.34) can be written in discretized form as follows,

$$\sigma_{kl} = \sum_{j=1}^N \left\{ \int_{\Gamma_j} \mathbf{D}_{kl} \, d\Gamma \right\} \mathbf{p}^j - \sum_{j=1}^N \left\{ \int_{\Gamma_j} \mathbf{S}_{kl} \, d\Gamma \right\} \mathbf{u}^j + \sum_{s=1}^M \left\{ \int_{\Omega_s} \mathbf{D}_{kl} \mathbf{b} \, d\Omega \right\} \quad (4.35)$$

Integration

As we will see for the case of the quadratic elements, all integrals in the above expressions can be done numerically. For the case of constant elements however

it is simpler and more exact to carry out some integrations analytically, particularly those over the element with the singularity, i.e. for the case $i = j$. Everywhere else the values of the integrals in $\mathbf{H}^{ij}$ and $\mathbf{G}^{ij}$ have been computed using a four-point Gauss quadrature formula. Notice that values of the submatrices $\mathbf{H}^{ij}$ (for $i = j$) are easy to calculate using rigid body considerations as shown in 3.5. The terms in $\mathbf{G}^{ii}$ are the only coefficients to be computed analytically. The integrals are carried out for all the elements in $\mathbf{G}^{ii}$, i.e.

$$\mathbf{G}^{ii} = \begin{bmatrix} G_{11} & G_{12} \\ G_{21} & G_{22} \end{bmatrix}^{ii} \quad (4.36)$$

Substituting the fundamental solution into the corresponding integrals one can define three terms as,

$$G_{11} = \frac{1}{8\pi\mu(1-\nu)} \left[(3-4\nu) \int_{\Gamma_i} \ln\left(\frac{1}{r}\right) d\Gamma + \int_{\Gamma_i} \left(\frac{\partial r}{\partial x_1}\right)^2 d\Gamma \right] \quad (4.37)$$

$$G_{12} = G_{21} = \frac{1}{8\pi\mu(1-\nu)} \left[\int_{\Gamma_i} \frac{\partial r}{\partial x_1} \frac{\partial r}{\partial x_2} d\Gamma \right] \quad (4.38)$$

$$G_{22} = \frac{1}{8\pi\mu(1-\nu)} \left[(3-4\nu) \int_{\Gamma_i} \ln\left(\frac{1}{r}\right) d\Gamma + \int_{\Gamma_i} \left(\frac{\partial r}{\partial x_2}\right)^2 d\Gamma \right] \quad (4.39)$$

Notice that Γ_i refers to element 'i' over which the singularity is acting. The derivatives of r in formulae (4.37) to (4.39) are given, for the general case, by (figure 4.3(a)).

$$\begin{aligned} \frac{\partial r}{\partial x_1} &= \frac{r_1}{r} = \frac{x_1^k - x_1^i}{|r|} \\ \frac{\partial r}{\partial x_2} &= \frac{r_2}{r} = \frac{x_2^k - x_2^i}{|r|} \end{aligned} \quad (4.40)$$

where $|r|$ is the magnitude of the distance vector $\vec{r}$.

With the above definitions, one can now concentrate on the case for which the integration is carried out over the element 'i' which contains the singularity (figure 4.3(b)). Let us consider that the element starts at extreme point (1) and finishes at extreme point (2) shown in figure 4.3(a). The distances from these points to the singularity located at i in the centre of the element is R , while small r is the variable distance from i to any point over the element, i.e. r is equivalent to Γ .

Using the $\theta - r$ system defined in figure 4.3(b) one finds the following relationship for (4.40)

$$\begin{aligned} \frac{\partial r}{\partial x_1} &= \frac{r_1}{r} = \cos \theta \\ \frac{\partial r}{\partial x_2} &= \frac{r_2}{r} = \sin \theta \end{aligned} \quad (4.41)$$

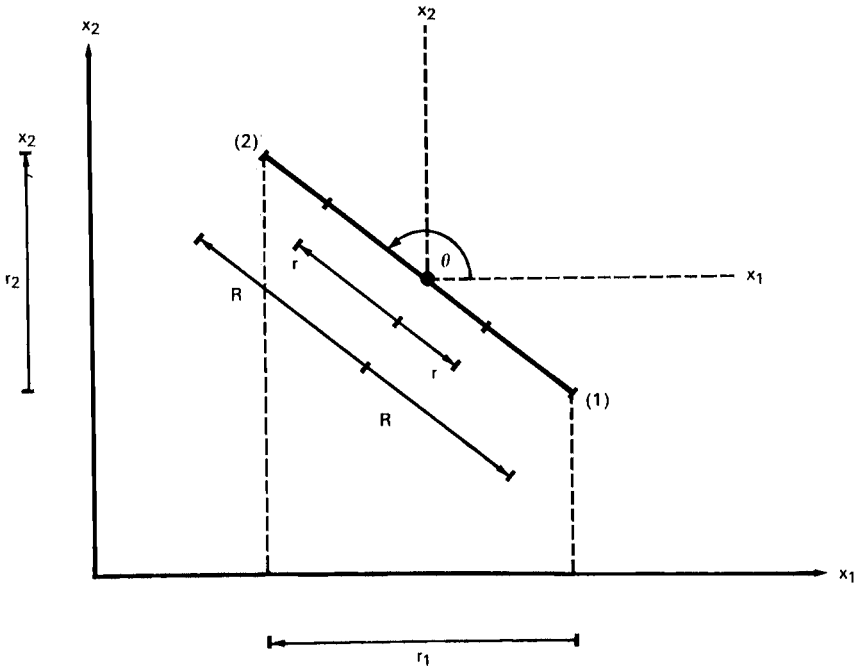
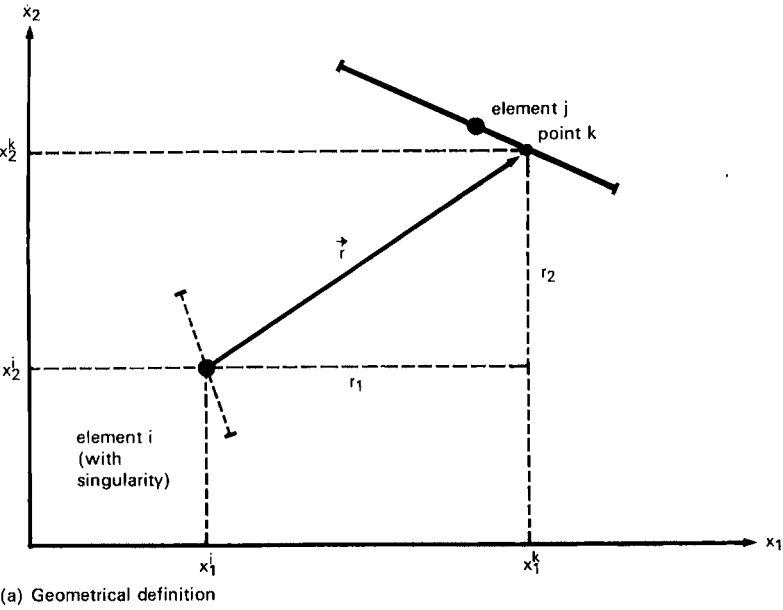


Figure 4.3

One can now write formulae (4.37) to (4.39) taking limits around the singularity. (Notice that the integrals from i to (1) or (2) are the same.)

$$\begin{aligned}
 G_{11} &= \lim_{\epsilon \rightarrow 0} \left\{ \frac{1}{8\pi\mu(1-\nu)} \left[2(3-4\nu) \int_{\epsilon}^R \ln \frac{1}{r} dr + 2 \int_{\epsilon}^R \cos^2 \theta dr \right] \right\} \\
 &= \lim_{\epsilon \rightarrow 0} \left\{ \frac{1}{8\pi\mu(1-\nu)} \left[2(3-4\nu) \left\{ \left[r \ln \frac{1}{r} \right]_{\epsilon}^R + [r]_{\epsilon}^R \right\} + 2R \cos^2 \theta \right] \right\} \\
 &= \frac{2R}{8\pi\mu(1-\nu)} \left[(3-4\nu)(1 - \ln R) + \frac{(r_1)^2}{4R^2} \right] \quad (4.42)
 \end{aligned}$$

$$\begin{aligned}
 G_{12} = G_{21} &= \lim_{\epsilon \rightarrow 0} \left\{ \frac{1}{8\pi\mu(1-\nu)} \left[2 \int_{\epsilon}^R \sin \theta \cos \theta d\Gamma \right] \right\} \\
 &= \frac{1}{8\pi\mu(1-\nu)} [2R \sin \theta \cos \theta] = \frac{2R}{8\pi\mu(1-\nu)} \left[\frac{r_1 r_2}{4R^2} \right] \quad (4.43)
 \end{aligned}$$

$$\begin{aligned}
 G_{22} &= \lim_{\epsilon \rightarrow 0} \left\{ \frac{1}{8\pi\mu(1-\nu)} \left[2(3-4\nu) \int_{\epsilon}^R \ln \frac{1}{r} dr + 2 \int_{\epsilon}^R \sin^2 \theta dr \right] \right\} \\
 &= \lim_{\epsilon \rightarrow 0} \left\{ \frac{1}{8\pi\mu(1-\nu)} \left[2(3-4\nu) \left\{ \left[r \ln \frac{1}{r} \right]_{\epsilon}^R + [r]_{\epsilon}^R \right\} + 2R \sin^2 \theta \right] \right\} \\
 &= \frac{2R}{8\pi\mu(1-\nu)} \left[(3-4\nu)(1 - \ln R) + \frac{(r_2)^2}{4R^2} \right] \quad (4.44)
 \end{aligned}$$

4.5 Elastostatics Code using Constant Elements (ELCONBE)

This section describes a computer code for the solution of two dimensional isotropic elastostatics problems without body forces. The code has a similar organization as those described in Chapter 2 and because of this the routines are given names similar to those used previously. The code is written for plane strain but can be used for plate stretching as well, simply by giving an equivalent value of ν (notice that the shear modulus μ or G remains the same).

The variables used in the program and their meaning are listed below,

- N: Number of boundary elements (equal to number of nodes for this case).
- L: Number of internal points where the displacements and stresses are to be calculated.
- M: Number of different surfaces (maximum 5).
- NC(): Number of the last node of each different surface.
- GE: Shear modulus.
- XNU: Poisson's ratio for plane strain; use a fictitious Poisson's ratio $XNU = \nu/(1 + \nu)$ for plane stress, ν being the Poisson's ratio.

- X,Y: One dimensional arrays with x_1 and x_2 coordinates of the extreme points of the boundary elements.
- XM,YM: One dimensional arrays with the coordinates of the nodes.
- G: Matrix defined in equation (4.30). After application of the boundary conditions the matrix A is stored in the same location (equation (4.31)).
- H: Matrix defined by equation (4.30).
- KODE: One dimensional array indicating the type of boundary condition at the element nodes. KODE = 0 means that a displacement is prescribed and KODE = 1 that a traction is prescribed.
- FI: Vector where the prescribed values of boundary conditions are stored. Each element is associated with a value of KODE.
- DFI: Right hand side vector in the global system. After solution it contains the values of the unknowns.
- CX,CY: One dimensional vectors with coordinates of internal points (maximum number 20).
- DSOL: Values of displacements at internal points (2 displacements per point).
- SSOL: Values of stresses at internal points (3 stresses per point).

Body forces have not been introduced for simplicity.

1. Main Program

The FORTRAN listing for the main program is as follows:

```

C-----
C
C      PROGRAM ELCONBE
C
C      PROGRAM 28
C
C      THIS PROGRAM SOLVES TWO-DIMENSIONAL (EL)ASTIC PROBLEMS
C      USING (CON)STANT (B)OUNDARY (E)LEMENTS
C
C      CHARACTER*10 FILEIN,FILEOUT
C
C      COMMON/MATG/ G(100,100)
C      COMMON/MATH/ H(100,100)
C      COMMON N,L,NC(5),M,GE,XNU,INP,IPR
C      DIMENSION X(51),Y(51),XM(50),YM(50),FI(100),DFI(100)
C      DIMENSION KODE(100),CX(20),CY(20),SSOL(60),DSOL(40)
C
C      SET MAXIMUN DIMENSION OF THE SYSTEM OF EQUATIONS (NX)
C      (THIS NUMBER MUST BE EQUAL OR SMALLER THAN THE DIMENSION OF G AND H)
C
C      NX=100
C
C      ASSIGN NUMBERS FOR INPUT AND OUTPUT FILES
C
C      INP=5
C      IPR=6
C
C      READ NAMES AND OPEN FILES FOR INPUT AND UOTPUT
C
C      WRITE(*,' (A) ') ' NAME OF THE INPUT FILE (MAX. 10 CHART.) '
C      READ(*,' (A) ') FILEIN
C      OPEN(INP,FILE=FILEIN,STATUS='OLD')

```

```

      WRITE(*,' (A) ') ' NAME OF THE OUTPUT FILE (MAX.10 CHART.) '
      READ(*,' (A) ') FILEOUT
      OPEN(IPR,FILE=FILEOUT,STATUS='NEW')
C
C  READ DATA
C
      CALL INPUTEC(CX,CY,X,Y,KODE,FI)
C
C  COMPUTE H AND G MATRICES AND FORM SYSTEM (A X = F)
C
      CALL GHMATEC(X,Y,XM,YM,G,H,FI,DFI,KODE,NX)
C
C  SOLVE SYSTEM OF EQUATIONS
C
      NN=2*N
      CALL SLNPD(G,DFI,D,NN,NX)
C
C  COMPUTE STRESS AND DISPLACEMENT AT INTERNAL POINTS
C
      CALL INTEREC(FI,DFI,KODE,CX,CY,X,Y,SSOL,DSOL)
C
C  PRINT RESULTS AT BOUNDARY NODES AND INTERNAL POINTS
C
      CALL OUTPTEC(XM,YM,FI,DFI,CX,CY,SSOL,DSOL)
      CLOSE (INP)
      CLOSE (IPR)
      STOP
      END

```

2. Routine INPUTEC

This subroutine reads the input required by the code. A file whose name is requested from the user by the main program will contain the following lines of free format input.

- (i) *Title Line* Contains the title of the problem.
- (ii) *Basic parameter line* Contains the number of elements, number of surfaces, last nodes of each different surface, the shear modulus and Poisson ratio or fictitious Poisson ratio for plane stress problems $\nu/(1 + \nu)$.
- (iii) *Extreme Points of Boundary Elements Lines* The coordinates of the extreme point of an element are read in the counterclockwise direction for external surfaces and clockwise for internal ones. The coordinates are read in free FORMAT.
- (iv) *Boundary Conditions Lines* As many lines as boundary nodes giving the values of the known variable in x_1 and x_2 directions. The variables are displacements if $KODE = 0$ or tractions if $KODE = 1$.
- (v) *Internal Points Coordinate Lines* Internal nodes are defined, each with the x_1, x_2 coordinates of the point. The coordinates are read in free FORMAT.

The subroutine prints first the name of the job and the basic parameters. Then the coordinates of the extreme points of the elements and the boundary conditions given by node number, codes and prescribed values are printed. The internal point coordinates will be printed in the subroutine OUTPTEC. Its listing is as follows:

```

C-----
      SUBROUTINE INPUTEC(CX,CY,X,Y,KODE,FI)
C
C  PROGRAM 29
C

```

```

      CHARACTER*80 TITLE
      DIMENSION CX(1),CY(1),X(1),Y(1),KODE(1),FI(1)
      COMMON N,L,NC(5),M,GE,XNU,INP,IPR
C
C   N= NUMBER OF BOUNDARY NODES (= NUMBER OF ELEMENTS)
C   L= NUMBER OF INTERNAL POINTS WHERE DISPLACEMENT AND STRESS
C   ARE CALCULATED
C   M= NUMBER OF DIFFERENT BOUNDARIES
C   NC(I)= LAST NODE OF BOUNDARY I
C   GE= SHEAR MODULUS
C   XNU= POISSON MODULUS
C
      WRITE(IPR,100)
100  FORMAT(/' ',79('*'))
C
C   READ JOB TITLE
C
      READ(INP,'(A)') TITLE
      WRITE(IPR,'(A)') TITLE
C
C   READ NUMBER OF NODES, INTERNAL POINTS AND DIFFERENT BOUNDARIES;
C   READ LAST NODES OF THESE BOUNDARIES AND MATERIAL PROPERTIES
C
      READ(INP,*)N,L,M,(NC(K),K=1,5),GE,XNU
      WRITE(IPR,300)N,L,GE,XNU
300  FORMAT(/' DATA'//2X,'NUMBER OF BOUNDARY ELEMENTS =' ,I3/2X,
1'NUMBER OF INTERNAL POINTS =' ,I3/2X,
2'SHEAR MODULUS =' ,E14.7/2X,'POISSON RATIO =' ,E14.7)
      IF(M)40,40,30
30  WRITE(IPR,999)M,(NC(K),K=1,M)
999  FORMAT(2X,'NUMBER OF DIFFERENT BOUNDARIES=' ,I3/2X,
1'LAST NODES OF THESE BOUNDARIES =' ,5(2X,I3))
C
C   READ COORDINATES OF EXTREME POINTS OF THE BOUNDARY
C   ELEMENTS IN ARRAYS X AND Y
C
40  WRITE(IPR,500)
500  FORMAT(/2X,'COORDINATES OF THE EXTREME POINTS OF'
1' THE BOUNDARY ELEMENTS'//4X,'POINT',10X,'X',18X,'Y')
      READ(INP,*) (X(I),Y(I),I=1,N)
      DO 10 I=1,N
10  WRITE(IPR,700)I,X(I),Y(I)
700  FORMAT(5X,I3,2(5X,E14.7))
C
C   READ BOUNDARY CONDITIONS IN FI(I) VECTOR, IF KODE(I)=0 THE FI(I)
C   VALUE IS A KNOWN DISPLACEMENT; IF KODE(I)=1 THE FI(I) VALUE IS A
C   KNOWN TRACTION.
C
      WRITE(IPR,800)
800  FORMAT(/2X,'BOUNDARY CONDITIONS'//15X,'PRESCRIBED VALUE',15X,
1'PRESCRIBED VALUE'/5X,'NODE',9X,'X DIRECTION',8X,'CODE',8X,
2'Y DIRECTION',8X,'CODE')
      DO 20 I=1,N
      READ(INP,*) KODE(2*I-1),FI(2*I-1),KODE(2*I),FI(2*I)
20  WRITE(IPR,950)I,FI(2*I-1),KODE(2*I-1),FI(2*I),KODE(2*I)
950  FORMAT(5X,I3,8X,E14.7,8X,I1,8X,E14.7,8X,I1)
C
C   READ COORDINATES OF THE INTERNAL POINTS
C
      IF(L.EQ.0) GO TO 50
      READ(INP,*) (CX(I),CY(I),I=1,L)
50  RETURN
      END

```

3. Routine GHMATEC

This routine computes the system matrices **H** and **G** by calling the routines EXTINEC and LOCINEC, i.e.

- (i) *EXTINEC* Computes the $\mathbf{H}^{ij}$ and $\mathbf{G}^{ij}$ submatrices using numerical integration for the cases $i \neq j$.
- (ii) *LOCINEC* Only calculates the submatrices $\mathbf{G}^{ii}$ applying formulae (4.42) to (4.44).

Notice that the diagonal submatrices of $\mathbf{H}$ for a smooth boundary as in constant elements, are given by

$$\mathbf{H}^{ii} = \begin{bmatrix} \frac{1}{2} & 0 \\ 0 & \frac{1}{2} \end{bmatrix}$$

After computing $\mathbf{H}$ and $\mathbf{G}$ matrices the routine arranges the equations in accordance with the boundary conditions and forms the $\mathbf{A}$ matrix of equations (4.31). When rearranging the columns those of $\mathbf{G}$ are multiplied by the shear modulus (called *GE* in the code) so that all elements of $\mathbf{A}$ are of the same order. This helps to increase the numerical accuracy of the solution.

The right hand side vector $\mathbf{F}$ of equation (4.31) also found in this routine is stored in array *DFI*.

GHMATEC listing is as follows:

```

C-----
      SUBROUTINE GHMATEC(X,Y,XM,YM,G,H,FI,DFI,KODE,NX)
C
C   PROGRAM 30
C
C   THIS SUBROUTINE COMPUTES THE G AND H MATRICES AND
C   FORMS THE SYSTEM OF EQUATIONS A X = F
C
      DIMENSION G(NX,NX),H(NX,NX)
      DIMENSION X(1),Y(1),XM(1),YM(1),FI(1)
      DIMENSION KODE(1),DFI(1)
      COMMON N,L,NC(5),M,GE,XNU,INP,IPR
C
C   COMPUTE THE NODAL COORDINATES AND STORE IN ARRAYS XM AND YM
C
      X(N+1)=X(1)
      Y(N+1)=Y(1)
      DO 10 I=1,N
        XM(I)=(X(I)+X(I+1))/2
10     YM(I)=(Y(I)+Y(I+1))/2
        IF(M-1)15,15,12
12     XM(NC(1))=(X(NC(1))+X(1))/2
        YM(NC(1))=(Y(NC(1))+Y(1))/2
        DO 13 K=2,M
          XM(NC(K))=(X(NC(K))+X(NC(K-1)+1))/2
13     YM(NC(K))=(Y(NC(K))+Y(NC(K-1)+1))/2
C
C   COMPUTE THE COEFFICIENTS OF G AND H MATRICES
C
15     DO 30 I=1,N
16     DO 30 J=1,N
17       IF(M-1)16,16,17
18       KK=1
19       GO TO 23
20       IF(J-NC(K))22,21,22
21       KK=NC(K-1)+1
22       GO TO 23
23     CONTINUE
16     KK=J+1
23     IF(I-J)20,25,20

```

```

20 CALL EXTINEC(XM(I),YM(I),X(J),Y(J),X(KK),Y(KK),H((2*I-1),(2*J-1)),
  1H((2*I-1),(2*J)),H((2*I),(2*J-1)),H((2*I),(2*J)),G((2*I-1),
  2(2*J-1)),G((2*I-1),(2*J)),G((2*I),(2*J)))
  G((2*I),(2*J-1))=G((2*I-1),(2*J))
  GO TO 26
25 CALL LOCINEC(X(J),Y(J),X(KK),Y(KK),G((2*I-1),(2*J-1)),
  1G((2*I-1),(2*J)),G((2*I),(2*J)))
  H((2*I-1),(2*J-1))=0.5
  H((2*I),(2*J))=0.5
  H((2*I-1),(2*J))=0.
  H((2*I),(2*J-1))=0.
  G((2*I),(2*J-1))=G((2*I-1),(2*J))
26 CONTINUE
30 CONTINUE

C
C REORDER THE COLUMNS OF THE SYSTEM OF EQUATIONS IN ACCORDANCE
C WITH THE BOUNDARY CONDITIONS AND FORM SYSTEM MATRIX A WHICH
C IS STORED IN G
C
  NN=2*N
  DO 50 J=1,NN
    IF(KODE(J))43,43,40
40 DO 42 I=1,NN
    CH=G(I,J)
    G(I,J)=-H(I,J)
42 H(I,J)=-CH
    GO TO 50
43 DO 45 I=1,NN
45 G(I,J)=G(I,J)*GE
50 CONTINUE

C
C FORM THE RIGHT HAND SIDE VECTOR F WHICH IS STORED IN DFI
C
  DO 60 I=1,NN
    DFI(I)=0.
    DO 60 J=1,NN
      DFI(I)=DFI(I)+H(I,J)*FI(J)
60 CONTINUE
  RETURN
  END

```

4. Routine EXTINEC

This routine computes the G^{ij} and H^{ij} submatrices (for $i \neq j$) using numerical integration. Notice that G^{ij} is symmetric but H^{ij} is not. The values of the coefficients are computed using four points Gaussian quadrature.

```

C-----
  SUBROUTINE EXTINEC(XP,YP,X1,Y1,X2,Y2,H11,H12,H21,H22,G11,G12,G22)
C
C PROGRAM 31
C
C THIS SOUBROUTINE COMPUTES THE G AND H MATRICES
C COEFFICIENTS THAT RELATE A COLOCATION POINT WITH A DIFFERENT
C ELEMENT USING GAUSS QUADRATURE
C
C DIST= DISTANCE FROM THE COLOCATION POINT TO THE
C LINE TANGENT TO THE ELEMENT
C RA= DISTANCE FROM THE COLOCATION POINT TO THE
C GAUSS INTEGRATION POINT AT THE BOUNDARY ELEMENT
C
  DIMENSION XCO(4),YCO(4),GI(4),OME(4)
  COMMON N,L,NC(5),M,GE,XNU,INP,IPR
  DATA GI/0.86113631,-0.86113631,0.33998104,-0.33998104/
  DATA OME/0.34785485,0.34785485,0.65214515,0.65214515/
C
  AX=(X2-X1)/2
  BX=(X2+X1)/2

```

```

      AY=(Y2-Y1)/2
      BY=(Y2+Y1)/2
      ETA1=(Y2-Y1)/(2*SQRT(AX**2+AY**2))
      ETA2=(X1-X2)/(2*SQRT(AX**2+AY**2))
C
C  COMPUTE THE DISTANCE FROM THE POINT TO THE LINE OF THE ELEMENT
C
      IF(AX) 10,20,10
10  TA=AY/AX
      DIST=ABS((TA*XP-YP+Y1-TA*X1)/SQRT(TA**2+1))
      GO TO 30
20  DIST=ABS(XP-X1)
C
C  DETERMINE THE DIRECTION OF THE OUTWARD NORMAL
C
30  SIG=(X1-XP)*(Y2-YP)-(X2-XP)*(Y1-YP)
      IF(SIG) 31,32,32
31  DIST=-DIST
32  H11=0.
      H12=0.
      H21=0.
      H22=0.
      G11=0.
      G12=0.
      G22=0.
C
C  COMPUTE G AND H COEFFICIENTS
C
      DE=4*3.141592*(1-XNU)
      DO 40 I=1,4
      XCO(I)=AX*GI(I)+BX
      YCO(I)=AY*GI(I)+BY
      RA=SQRT((XP-XCO(I))**2+(YP-YCO(I))**2)
      RD1=(XCO(I)-XP)/RA
      RD2=(YCO(I)-YP)/RA
      G11=G11+((3-4*XNU)*ALOG(1./RA)+RD1**2)*OME(I)*SQRT(AX**2+AY**2)/(2
1*DE*GE)
      G12=G12+RD1*RD2*OME(I)*SQRT(AX**2+AY**2)/(2*DE*GE)
      G22=G22+((3-4*XNU)*ALOG(1./RA)+RD2**2)*OME(I)*SQRT(AX**2+AY**2)/(2
1*DE*GE)
      H11=H11-DIST*((1-2*XNU)+2*RD1**2)/(RA**2*DE)*OME(I)*SQRT(AX**2+AY*
1*2)
      H12=H12-(DIST*2*RD1*RD2/RA+(1-2*XNU)*(ETA1*RD2-ETA2*RD1))*OME(I)*S
1QRT(AX**2+AY**2)/(RA*DE)
      H21=H21-(DIST*2*RD1*RD2/RA+(1-2*XNU)*(ETA2*RD1-ETA1*RD2))*OME(I)*S
1QRT(AX**2+AY**2)/(RA*DE)
40  H22=H22-DIST*((1-2*XNU)+2*RD2**2)*OME(I)*SQRT(AX**2+AY**2)/(RA**2*
1DE)
      RETURN
      END

```

5. Routine LOCINEC

It calculates the G^{ii} submatrix in accordance with formulae (4.42) to (4.44), which give the analytical results.

```

C-----
      SUBROUTINE LOCINEC(X1,Y1,X2,Y2,G11,G12,G22)
C
C  PROGRAM 32
C
C  THIS SUBROUTINE COMPUTES THE VALUES OF THE MATRIX G COEFFICIENTS
C  THAT RELATE AN ELEMENT WITH ITSELF
C
      COMMON N,L,NC(5),M,GE,XNU,INP,IPR
C
      AX=(X2-X1)/2
      AY=(Y2-Y1)/2
      SR=SQRT(AX**2+AY**2)

```

```

DE=4*3.141592*GE*(1-XNU)
G11=SR*((3-4*XNU)*(1-ALOG(SR))+(X2-X1)**2/(4*SR**2))/DE
G22=SR*((3-4*XNU)*(1-ALOG(SR))+(Y2-Y1)**2/(4*SR**2))/DE
G12=(X2-X1)*(Y2-Y1)/(4*SR*DE)
RETURN
END

```

6. Routine SLNPD

This is the same solver as used in previous codes.

7. Routine INTEREC

This subroutine computes the displacements and stresses at internal points. First it reorders the vectors DFI and FI. Notice that the first contains the values of X in equation (4.31) as given by the solver and the second the boundary conditions. After reorganization the boundary displacements are all stored in FI and the tractions in DFI.

Formulae (4.33) and (4.35) are applied to compute internal stresses and displacements. This requires the integration of terms with S and D coefficients which are carried out numerically in a routine called SIGMAEC.

The FORTRAN Listing is as follows:

```

C-----
      SUBROUTINE INTEREC(FI,DFI,KODE,CX,CY,X,Y,SSOL,DSOL)
C
C   PROGRAM 33
C
C   THIS SUBROUTINE COMPUTES THE VALUES OF THE STRESS AND DISPLACEMENT
C   COMPONENTS AT INTERNAL POINTS
C
      DIMENSION CX(1),CY(1),SSOL(1),DSOL(1)
      DIMENSION FI(1),DFI(1),KODE(1),X(1),Y(1)
      COMMON N,L,NC(5),M,GE,XNU,INP,IPR
C
C   REARRANGE FI AND DFI ARRAYS TO STORE ALL THE VALUES
C   OF THE DISPLACEMENT IN FI AND ALL THE VALUES OF THE TRACTIONS IN DFI
C
      NN=2*N
      DO 20 I=1,NN
        IF(KODE(I)) 15,15,10
      10 CH=FI(I)
        FI(I)=DFI(I)
        DFI(I)=CH
        GO TO 20
      15 DFI(I)=DFI(I)*GE
      20 CONTINUE
C
C   COMPUTE THE VALUES OF STRESSES AND DISPLACEMENTS
C   AT INTERNAL POINTS.
C
      IF(L.EQ.0) GO TO 50
      DO 40 K=1,L
        DSOL(2*K-1)=0.
        DSOL(2*K)=0.
        SSOL(3*K-2)=0.
        SSOL(3*K-1)=0.
        SSOL(3*K)=0.
        DO 30 J=1,N
          IF(M-1) 28,28,22
        22 IF(J-NC(1)) 24,23,24

```

```

23 KK=1
   GO TO 29
24 DO 26 LK=2,M
   IF(J-NC(LK))26,25,26
25 KK=NC(LK-1)+1
   GO TO 29
26 CONTINUE
28 KK=J+1
29 CALL EXTINEC(CX(K),CY(K),X(J),Y(J),X(KK),Y(KK),H11,H12,H21,H22,
  1 G11,G12,G22)
   DSOL(2*K-1)=DSOL(2*K-1)+DFI(2*J-1)*G11+DFI(2*J)*G12-FI(2*J-1)*H11-
  1FI(2*J)*H12
   DSOL(2*K)=DSOL(2*K)+DFI(2*J-1)*G12+DFI(2*J)*G22-FI(2*J-1)*H21-FI(2
  1*J)*H22
   CALL SIGMAEC(CX(K),CY(K),X(J),Y(J),X(KK),Y(KK),D111,D211,D112,
  1 D212,D122,D222,S111,S211,S112,S212,S122,S222)
   SSOL(3*K-2)=SSOL(3*K-2)+DFI(2*J-1)*D111+DFI(2*J)*D211-FI(2*J-1)*S1
  111-FI(2*J)*S211
   SSOL(3*K-1)=SSOL(3*K-1)+DFI(2*J-1)*D112+DFI(2*J)*D212-FI(2*J-1)*S1
  112-FI(2*J)*S212
  30 SSOL(3*K)=SSOL(3*K)+DFI(2*J-1)*D122+DFI(2*J)*D222-FI(2*J-1)*S122-F
  1I(2*J)*S222
  40 CONTINUE
  50 RETURN
   END

```

8. Routine SIGMAEC

This subroutine computes the integrals with S and D terms described in formulae (4.35) and needed to calculate the internal stresses. The integrations are carried out using a four points Gaussian integration scheme.

```

C-----
      SUBROUTINE SIGMAEC(XP,YP,X1,Y1,X2,Y2,D111,D211,D112,D212,D122,
  1 D222,S111,S211,S112,S212,S122,S222)
C
C  PROGRAM 34
C
C  THIS SUBROUTINE COMPUTES THE VALUES OF THE S AND D MATRICES
C  USING GAUSS QUADRATURE IN ORDER TO COMPUTE THE STRESSES
C  AT ANY INTERNAL POINT
C
C  RA= DISTANCE FROM THE POINT TO THE GAUSS INTEGRATION POINTS
C      ON THE BOUNDARY ELEMENTS
C  DIST= DISTANCE FROM THE POINT TO THE LINE TANGENT
C        TO THE ELEMENT
C  RD1,RD2= DERIVATIVES OF RA
C  ETA1,ETA2= COMPONENTS OF THE UNIT NORMAL TO THE ELEMENT
C
      DIMENSION XCO(4),YCO(4),GI(4),OME(4)
      COMMON N,L,NC(5),M,GE,XNU,INP,IPR
      DATA GI/0.86113631,-0.86113631,0.33998104,-0.33998104/
      DATA OME/0.34785485,0.34785485,0.65214515,0.65214515/
C
      AX=(X2-X1)/2
      BX=(X2+X1)/2
      AY=(Y2-Y1)/2
      BY=(Y2+Y1)/2
      ETA1=(Y2-Y1)/(2*SQRT(AX**2+AY**2))
      ETA2=(X1-X2)/(2*SQRT(AX**2+AY**2))
C
C  COMPUTE THE DISTANCE FROM THE POINT TO THE LINE OF THE ELEMENT
C
      IF(AX)10,20,10
  10 TA=AY/AX
      DIST=ABS((TA*XP-YP+Y1-TA*X1)/SQRT(TA**2+1))
      GO TO 30
  20 DIST=ABS(XP-X1)

```

```

C
C DETERMINE THE DIRECTION OF THE OUTWARD NORMAL
C
30 SIG=(X1-XP)*(Y2-YP)-(X2-XP)*(Y1-YP)
    IF (SIG) 31,32,32
31 DIST=-DIST
32 D111=0.
    D211=0.
    D112=0.
    D212=0.
    D122=0.
    D222=0.
    S111=0.
    S211=0.
    S112=0.
    S212=0.
    S122=0.
    S222=0.

C
C COMPUTE D AND S COEFFICIENTS
C
    FA=1-4*XNU
    AL=1-2*XNU
    DE=4*3.141592*(1-XNU)
    DO 40 I=1,4
        XCO(I)=AX*GI(I)+BX
        YCO(I)=AY*GI(I)+BY
        RA=SQRT((XP-XCO(I))**2+(YP-YCO(I))**2)
        RD1=(XCO(I)-XP)/RA
        RD2=(YCO(I)-YP)/RA
        D111=D111+(AL*RD1+2*RD1**3)*OME(I)*SQRT(AX**2+AY**2)/(DE*RA)
        D211=D211+(2*RD1**2*RD2-AL*RD2)*OME(I)*SQRT(AX**2+AY**2)/(DE*RA)
        D112=D112+(AL*RD2+2*RD1**2*RD2)/(DE*RA)*OME(I)*SQRT(AX**2+AY**2)
        D212=D212+(AL*RD1+2*RD1*RD2**2)/(DE*RA)*OME(I)*SQRT(AX**2+AY**2)
        D122=D122+(2*RD1*RD2**2-AL*RD1)/(DE*RA)*OME(I)*SQRT(AX**2+AY**2)
        D222=D222+(AL*RD2+2*RD2**3)/(DE*RA)*OME(I)*SQRT(AX**2+AY**2)
        S111=S111+(2*DIST/RA*(AL*RD1+XNU*2*RD1-4*RD1**3)+4*XNU*ETA1*RD1**2
1+AL*(2*ETA1*RD1**2+2*ETA1)-FA*ETA1)*2*GE/(DE*RA**2)*OME(I)*SQRT(AX
2**2+AY**2)
        S211=S211+(2*DIST/RA*(AL*RD2-4*RD1**2*RD2)+4*XNU*ETA1*RD1*RD2+AL*2
1*ETA2*RD1**2-FA*ETA2)*2*GE/(DE*RA**2)*OME(I)*SQRT(AX**2+AY**2)
        S112=S112+(2*DIST/RA*(XNU*RD2-4*RD1**2*RD2)+2*XNU*(ETA1*RD2*RD1+ET
1A2*RD1**2)+AL*(2*ETA1*RD1*RD2+ETA2))*2*GE/(DE*RA**2)*OME(I)*SQRT(A
2X**2+AY**2)
        S212=S212+(2*DIST/RA*(XNU*RD1-4*RD1*RD2**2)+2*XNU*(ETA1*RD2**2+ETA
12*RD1*RD2)+AL*(2*ETA2*RD1*RD2+ETA1))*2*GE/(DE*RA**2)*OME(I)*SQRT(A
2X**2+AY**2)
        S122=S122+(2*DIST/RA*(AL*RD1-4*RD1*RD2**2)+4*XNU*ETA2*RD1*RD2+AL*2
1*ETA1*RD2**2-FA*ETA1)*2*GE/(DE*RA**2)*OME(I)*SQRT(AX**2+AY**2)
40 S222=S222+(2*DIST/RA*(AL*RD2+2*XNU*RD2-4*RD2**3)+4*XNU*ETA2*RD2**2
1+AL*(2*ETA2*RD2**2+2*ETA2)-FA*ETA2)*2*GE/(DE*RA**2)*OME(I)*SQRT(AX
2**2+AY**2)
    RETURN
    END

```

9. Routine OUTPTEC

This subroutine prints the results in the following order.

- (ii) Boundary nodes values with coordinates x_1x_2 , values of u_1u_2 displacements and p_1p_2 tractions.
- (ii) Internal node results with coordinates x_1x_2 , values of u_1u_2 displacements and $\sigma_{11}\sigma_{12}\sigma_{22}$ stresses.

Notice that stresses on the boundary are not produced. If required they will need to be computed using formulae (3.5) with the derivatives of displacements

tangential to the surface computed by considering the nodes of displacements and adjacent elements and applying a finite different type approach.

The listing of OUTPT is as follows:

```

C-----
      SUBROUTINE OUTPTEC(XM,YM,FI,DFI,CX,CY,SSOL,DSOL)
C
C  PROGRAM 35
C
C  THIS SUBROUTINE PRINTS THE VALUES OF THE DISPLACEMENTS
C  AND TRACTIONS AT BOUNDARY NODES. IT ALSO PRINTS THE VALUES
C  OF DISPLACEMENTS AND STRESSES AT INTERNAL POINTS
C
      DIMENSION XM(1),YM(1),FI(1),DFI(1)
      DIMENSION CX(1),CY(1),SSOL(1),DSOL(1)
      COMMON N,L,NC(5),M,GE,XNU,INP,IPR
C
      WRITE(IPR,100)
100  FORMAT(' ',79('*'))//1X,'RESULTS'//2X,'BOUNDARY NODES'//6X
      1,'X',12X,'Y',9X,'DISPL. X',5X,'DISPL. Y',4X,
      2'TRACTION X',3X,'TRACTION Y'//)
      DO 10 I=1,N
10   WRITE(IPR,200) XM(I),YM(I),FI(2*I-1),FI(2*I),DFI(2*I-1),DFI(2*I)
200  FORMAT(6(1X,E12.5))
C
      IF(L.EQ.0.) GO TO 30
      WRITE(IPR,300)
300  FORMAT(/2X,'INTERNAL POINTS DISPLACEMENTS'//8X,'X',15X,'Y',10X,
      1'DISPLACEMENT X',5X,'DISPLACEMENT Y')
      DO 20 K=1,L
20   WRITE(IPR,400) CX(K),CY(K),DSOL(2*K-1),DSOL(2*K)
      WRITE(IPR,350)
350  FORMAT(/2X,'INTERNAL POINTS STRESSES'//8X,'X',15X,'Y',12X,
      1'SIGMA X',10X,'TAU XY',9X,'SIGMA Y')
      DO 25 K=1,L
25   WRITE(IPR,450) CX(K),CY(K),SSOL(3*K-2),SSOL(3*K-1),SSOL(3*K)
400  FORMAT(2(2X,E14.7),2(5X,E14.7))
450  FORMAT(5(2X,E14.7))
30   WRITE(IPR,500)
500  FORMAT(' ',79('*'))
      RETURN
      END

```

Example 4.1

Figure 4.4 describes a circular cavity under internal pressure in an infinite medium. Numerical results will be compared against the known analytical solution, first using constant elements and then with the quadratic elements of section 4.7 and 4.8.

The boundary is divided here into 24 constant elements and 5 internal nodes are defined at which displacements and stresses will be found.

Notice that in order to stop all rigid body movements one can suppress the displacements in the x_2 directions at node 18.

The input needed to run the ELCONBE code is as follows.

CIRCULAR CAVITY (DATA)

```

CIRCULAR CAVITY UNDER INTERNAL PRESSURE (24 CONSTANT ELEMENTS)
24 5 0 0 0 0 0 0 0 94500. 0.1
-.3916 -2.9743
-1.1481 -2.7716
-1.8263 -2.3801
-2.3801 -1.8263
-2.7716 -1.1480
-2.9743 -.3916
-2.9743 .3916

```